Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/05/2006
PI Name:	Flynn, Michael T. M.S.		
Project Title:	Advanced Development of the Direct Osmotic Concentration System		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mflynn@mail.arc.nasa.gov	**Fax:**	FY 650-604-1092
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1163
Organization Name:	NASA - Ames		
PI Address 1:	MS 239-15		
PI Address 2:			
PI Web Page:			
City:	Moffett FIeld	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Gormly, Sherwin (National Space Grant Foundation) Cath, Tzahi (University of Nevada, Reno) Childress, Amy (University of Nevada, Reno) Adams, Dean (University of Nevada Reno)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Direct osmotic concentration (DOC) is a highly integrated membrane treatment process for reclamation of space craft wastewater. It incorporates a novel forward osmosis (FO) process (previously referred to as direct osmosis), membrane evaporation process, reverse osmosis (RO) desalination, and a catalytic oxidation post-treatment step. As a result the DOC process has been selected for a NASA Rapid Technology Development Team (RTDT) effort. This project focuses on advancing the development of this technology to a level appropriate for human rated testing. A new prototype of a DOC system is been designed and fabricated that addresses the deficiencies encountered during the testing of the original system and able to achieve a long-term and reliable operation. This project will result in the development of a membrane processes for wastewater treatment that does not suffer from the common membrane problems of fouling and limited rejection of specific contaminants (e.g., urea, ammonia, surfactants, and various volatile compounds)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This technology will have limited Earth benefits as its objective is to convert human liquid wastes into drinking water. There is little demand for this on Earth. The component level technologies developed to accomplish this, however, may have applicability on Earth in a broad range of disciplines. (see www.hydrationtech.com). Membranes developed through this program are used in comercial water purification product for consumers and military.
Task Progress:	The work described herein represents the conclusion of the first year of this grant. Although some of the tasks were modified and several of the objectives of the original NRA were changed during this year, the work is essentially on schedule and within budget. All critical technological challenges have been met and a final design configuration has been agreed upon buy all members of the development team. Future work will focus on constructing a prototype and a final deliverable. At the beginning of this year several critical design issues were unresolved. The original DOC#1 module had proven to be unreliable due to continuing leakage. The DOC#2 module was also proving to be problematic and in the end proved to be unmanageable. The current configuration uses a new DOC#1 design and DCMD module in place of the original DOC#2 modules. At the beginning of the year the catalytic oxidation post-treatment system had not been tested and the DCMD heat pump had not been designed. Today, the catalytic polishing system is operational and the heat pump has been designed and components are being procured. Construction of the prototype system is about 40% complete. The DCMD module is functional, the FO system has been tested for integrity, and the RO system is essentially complete but as of yet untested. Within the next year, the prototype system will be completed and integrated testing will be initiated. Component level reliability testing will be completed and we will be transitioning into full scale prototype long duration testing. Construction of the final deliverable will be underway. All components will be procured and integration into the final deliverable will have been initiated. In the third year the prototype system will continue to be tested in a long duration testing program. The final deliverable will be completed and tested to verify performance. Any lessons learned from the long duration testing will be incorporated into the final design. By September 2007 a fully functional TRL 5-6 level DOC system will be delivered to NASA. The system will be capable of operating as an integrated water recycling system (urine, hygiene, humidity, and wash water) and in a transit only configuration (urine and humidity only). The transit configuration will be applicable to NASA’s proposed new Lunar Crew Exploration Vehicle (CEV) for all missions over 25 days in length. ESM calculations indicate that the DOC offers advantages for long duration missions. ESM guidelines for Lunar missions are not yet available. When they are we will conduct an analysis of the DOC within the Lunar constructs. Current ESM values indicate that the DOC is considerably less massive than the ISS WRS and roughly equivalent to the VPCAR system. Future will focus on reducing the ESM further.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Cath TY, Gormly S, Beaudry EG, Flynn MT, Adams VD, Childress AE. "Membrane contactor processes for wastewater reclamation in space: Part I. Direct osmotic concentration as pretreatment for reverse osmosis." Journal of Membrane Science, 2005 Jul 15;257(1-2):85-98. , Jul-2005
Articles in Peer-reviewed Journals	Cath TY, Adams VD, Childress AE. "Membrane contactor processes for wastewater reclamation in space: II. Combined direct osmosis, osmotic distillation, and membrane distillation for treatment of metabolic wastewater." Journal of Membrane Science, 2005 Jul 15;257(1-2):111-9. , Jul-2005
Papers from Meeting Proceedings	Cath TY, Gormly S, Beaudry EG, Flynn MT, Adams VD, Childress AE. "Progress in the Development of a Direct Osmotic Concentration Wastewater Recovery Process for Advanced Life Support Systems." International Conference on Environmental Systems, Rome, ITALY, July 2005. SAE paper, International Conference on Environmental Systems, July 2005. , Jul-2005
Papers from Meeting Proceedings	Gormly S, Adams VD, Cath TY, Childress AE, Flynn MT, Beaudry E. "Direct osmotic concentration: a primary water treatment process of space life support applications." 33rd International Conference on Environmental Systems, Vancouver, Canada, July 2003. SAE, ICES avalible at http://www.SAE.org , July 2003. , Jul-2003